

The Age of Misinformation: Older People Exhibit Greater Partisan Bias in Sharing and Evaluating (Mis)Information Accuracy

Guilherme A. Ramos^{1, 2} and Leaf Van Boven³

¹ Saunders College of Business, Rochester Institute of Technology

² Pensi Institute, Sao Paulo, Brazil

³ Department of Psychology and Neuroscience, University of Colorado Boulder

Older adults are known to share more political misinformation than younger adults, but the nature of this tendency remains unclear. Using signal detection theory, this research examines two explanations for older adults' greater political misinformation sharing: (a) Age reduces people's ability to discriminate true from false information, leading older people to share more misinformation, and (b) older people exhibit stronger partisan sharing, preferring to share information that is congruent with their political ingroup rather than the outgroup. Two exploratory studies around the 2022 presidential elections in Brazil and midterm elections in the United States ($N_{\text{Brazil}} = 697$; $N_{\text{U.S.}} = 1,716$) support the partisan congruence explanation. Age does not robustly predict discernment between true versus false information in sharing intentions or accuracy judgments. Instead, age predicts partisan biases in sharing political news and judging its accuracy, and these effects remain after controlling for income and education. The effects of age in predicting partisan information sharing are not statistically mediated by measures of cognitive reflection, need for cognition, or faith in intuition. These results suggest that older adults engage in more partisan information assessment and sharing behavior.

Public Significance Statement

Age is associated with greater sharing of political misinformation on social media. One potential explanation is that older individuals share more misinformation due to lower analytical tendencies, which impair their ability to discriminate true and false information. However, this research does not support lower truth discernment among older (vs. younger) adults for sharing intentions or for accuracy judgments. Instead, older adults' sharing intentions and accuracy judgments are consistent with increased partisan bias. People are more likely to share and to judge as true information that aligns with their political identities, regardless of its veracity, and this partisan bias increases with age. This suggests that older adults share more misinformation partly because they are more partisan.

Keywords: misinformation, age, partisan bias, signal detection theory, political thinking

Supplemental materials: <https://doi.org/10.1037/xge0001868.supp>

The spread of misinformation, including fabricated news stories, conspiracy theories, and hyperpartisan content, is a significant societal concern (The World Economic Forum, 2024). It erodes trust in democratic institutions and intensifies political polarization (Thagard, 2024; van der Linden, 2023). This issue is prevalent in democracies like the United States and Brazil, where misinformation has notably undermined confidence in such fundamental issues as election integrity (Berlinski et al., 2023; Rossini et al., 2023).

Age is an important predictor for misinformation sharing. Older adults are more likely to share misinformation than younger adults (Brashier & Schacter, 2020; Grinberg et al., 2019; Guess et al., 2019). For example, during the 2016 U.S. presidential election, Facebook

users over 65 shared almost seven times more fake news than adults under 30 (Grinberg et al., 2019). On Twitter, 80% of fake news was shared by users over 50 (Guess et al., 2019). The nature of older adults' misinformation sharing remains unclear, however. Given the projected growth of the older adult population in the coming decades (United Nations, Department of Economic and Social Affairs, Population Division, 2024), the growing tendency for people to consume content shared with them on social networks (Dixon, 2025), and the ease of sharing information, understanding what factors predict older adults' misinformation sharing is important both for basic psychological theory and for the broader impacts of such decisions.

This article was published Online First November 3, 2025.

Bolivar Reyes-Jaquez served as action editor.

Guilherme A. Ramos  <https://orcid.org/0000-0003-3519-3861>

Guilherme A. Ramos played a lead role in data curation, formal analysis, visualization, and writing—original draft and an equal role in conceptualization and methodology. Leaf Van Boven played a lead role in resources,

supervision, and writing—review and editing, a supporting role in writing—original draft, and an equal role in conceptualization and methodology.

Correspondence concerning this article should be addressed to Guilherme A. Ramos, Saunders College of Business, Rochester Institute of Technology, 105 Lomb Memorial Drive, Rochester, NY 14623, United States. Email: gramos@saunders.rit.edu

Building on theories of misinformation belief and sharing (Pennycook et al., 2021; Pennycook & Rand, 2019; Rathje et al., 2023; Van Bavel & Pereira, 2018) and using a signal detection theory approach (SDT; Gawronski et al., 2023; Green & Swets, 1966), the current research tests predictions derived from two potential explanations. First, older adults might be more susceptible to misinformation because they have a reduced ability to distinguish true from false information in their sharing decisions and accuracy assessments (Brashier & Schacter, 2020; Grinberg et al., 2019). According to this account, older adults have lower analytical thinking (Calso et al., 2020; Hess, 2014), which is a strong predictor of poor truth discernment (Pennycook & Rand, 2021). Lower analytical thinking may reduce truth discernment for older adults, and if they cannot discern true from false information, they may end up being less discriminating when sharing information. Lower analytical thinking might also be associated with a general tendency to share information, regardless of its veracity (Gawronski et al., 2023). Although there are findings suggestive of this explanation (Calso et al., 2020; Hess, 2014; Pehlivanoglu et al., 2022; Queen & Hess, 2010), other research, including a meta-analysis of 31 studies (Sultan et al., 2024), has provided direct evidence that older adults are actually better than younger adults at judging information accuracy (Allcott & Gentzkow, 2017; Arin et al., 2023; Huff & Umanath, 2018). These mixed findings for accuracy evaluations call for further examination of age, information discrimination, and analytical thinking, especially as they relate to sharing decisions.

The present article focuses on a second explanation for information sharing that centers on political identities and partisan information. People share political information that is congruent with their political identities. Republicans, for example, share information that reflects favorably on Republicans, if not unfavorably on Democrats (Gawronski et al., 2023). Compared with younger adults, older adults may share more misinformation partly because they exhibit a greater tendency to share information that is congruent with their partisan identity, whether the information is true or false. Older adults may simply be more partisan in their information sharing. Furthermore, older adults may not only exhibit a larger partisan bias in sharing information, but they may also exhibit a larger partisan bias in assessing the accuracy of information, believing in claims that align with their partisan identities.

The present article summarizes the results of two exploratory studies that support the identity-congruence explanation of age-related information sharing, while failing to provide evidence for the information discrimination and analytical thinking explanations of age-related misinformation sharing. The findings provide three contributions. First, although influential theories of misinformation sharing advance cognitive-based and motivational-based factors (Pennycook et al., 2021; Pennycook & Rand, 2019; Rathje et al., 2023; Van Bavel et al., 2024; Van Bavel & Pereira, 2018), there is a dearth of research on how those factors change with age. Second, we expand recent research investigating the relationship between age and accuracy assessments (Arin et al., 2023; Huff & Umanath, 2018; Sultan et al., 2024) by examining both sharing intentions (Studies 1 and 2) and accuracy assessments (Study 2). Finally, we conducted one study in both the United States and Brazil, two countries with starkly different political systems. While the United States is bipartisan, there were more than 30 political parties in Brazil during

our data collection period (Ramos et al., 2020, 2025). By probing the robustness of partisan dynamics in age-related information sharing across different institutional designs (Flores et al., 2022), this research also casts doubt on claims that politically polarizing processes are limited to the United States (Finkel et al., 2020).

Age and (Mis)Information Sharing

Although multiple factors contribute to misinformation sharing, scholarly debate has focused on two prevailing explanations (Martel et al., 2024; Pennycook et al., 2020, 2021; Pennycook & Rand, 2019; Rathje et al., 2023). The first emphasizes that older adults may have lower analytical reasoning, deliberation, and cognitive reflection (Pennycook et al., 2021; Pennycook & Rand, 2019). A tendency to avoid deliberation and reflection may lead people to exhibit incorrect judgments about information accuracy (Bago et al., 2020; Pennycook et al., 2021) and reduce discernment in sharing information (Gawronski et al., 2023).

There is suggestive evidence that older individuals can have reduced discernment, contributing to increased misinformation sharing. Age may reduce analytical thinking (Pehlivanoglu et al., 2022; Peters et al., 2007; Queen & Hess, 2010). Older adults exhibit memory loss and reduced processing, using their cognitive resources selectively (Hess, 2014; Hess & Ennis, 2012; Peters et al., 2007). This tendency could affect decision making particularly in situations that require deliberation (Hess et al., 2012), such as detecting misinformation. Older adults are also more vulnerable to false memories, source confusion, and prior exposure effects (Calso et al., 2020; Ebner et al., 2023; Henkel et al., 1998; Swire et al., 2017). Additionally, older adults often overlook sponsored content in advertising (Amazeen & Wojdyski, 2020), a tendency that in the context of misinformation can potentially further reduce their ability to detect misinformation.

Although these findings point to the possibility that age reduces discernment in accuracy judgments and sharing decisions, recent studies have provided evidence that older adults are not worse at judging information accuracy than their younger counterparts (Allcott & Gentzkow, 2017; Arin et al., 2023; Brashier & Schacter, 2020). A comprehensive meta-analysis of 31 studies found strong evidence that age increases truth discernment (Sultan et al., 2024). Thus, unless older adults are more likely than younger adults to intentionally share information that they believe to be false, which does not appear to be the case (Brashier & Schacter, 2020; Pennycook et al., 2018), it is unlikely older adults display reduced sharing discernment. Despite accumulating evidence against the negative effect of age on information discernment, an important contribution of the present studies is to replicate and extend such investigations in a new domain (sharing intentions) and in a country other than the United States (Brazil).

A second explanation for age-related misinformation sharing relates to older adults' partisan biases. According to this view, individuals are motivated to uphold their political ingroup identity (Van Bavel et al., 2024; Van Bavel & Pereira, 2018). Individuals are therefore more likely to believe in information that is congruent rather than incongruent with their partisan identities, regardless of whether the information is true (Lawson & Kakkar, 2022; Rathje et al., 2023). This tendency of upholding the political ingroup identity also leads individuals to privilege sharing information that is congruent with their partisan identities rather than information that is incongruent

(Gawronski et al., 2023). Prior work has investigated the roles of social identity and motivation reasoning in misinformation belief (Batailler et al., 2022; Nahon et al., 2024; Pennycook et al., 2022; Pennycook & Rand, 2019; Rathje et al., 2023; Sultan et al., 2022, 2024) and sharing (Gawronski et al., 2023). To our knowledge, however, previous research has not examined whether such political identity congruence effects vary with age.

Older adults may be particularly prone to partisan biases. Older adults are often more embedded in “filter bubbles” on social media, consistently receiving politically congruent information (Ghosh & Iyengar, 2019). They often have more entrenched political views than younger adults (Stoker & Jennings, 2008), thereby being less receptive to incongruent information than younger adults. Additionally, older adults experience greater affective polarization, characterized by stronger positive feelings toward their political ingroup and negative feelings toward outgroups (Phillips, 2022), which may exacerbate their motivation to protect their identity when evaluating and deciding whether to share information (Jenke, 2024). Consequently, older adults might preferentially share information aligned with their partisan identities, regardless of whether the information is true.

In addition to these two prevailing explanations, a hybrid explanation suggests that people may deploy their analytical thinking tendencies in ways that serve their partisan goals (Kahan et al., 2017; Kunda, 1990; Stanovich, 2021; Van Boven et al., 2019). Such individuals may use analytical thinking to be more discriminating of identity-incongruent information than identity-congruent information (Kahan et al., 2017; Kunda, 1990), and differential discrimination might be more pronounced among older adults. Thus, partisan bias in information sharing may be especially pronounced among older people with higher analytical thinking.

SDT and Misinformation

To disentangle these explanations for why older adults share more political misinformation than younger requires differentiating their truth discernment when deciding to share information from their overall tendency to share information, their discernment of true from false information, and their differential discernment of true versus false information that is congruent or incongruent with their partisan identities. SDT provides such an analytical framework (Batailler et al., 2022; Gawronski et al., 2023, 2024, 2025; Green & Swets, 1966; Nahon et al., 2024; Sultan et al., 2024). Although SDT has been used in misinformation research primarily to analyze accuracy judgments (e.g., Batailler et al., 2022; Sultan et al., 2024), prior work has also employed SDT to analyze sharing decisions (Gawronski et al., 2023). Our first study focuses on information sharing intentions as the primary outcome measure, while our second study includes decisions about both information sharing and veracity.

In an SDT framework applied to the context of misinformation, individuals evaluate multiple pieces of information that are either true or false. Then, they either (a) provide their truth assessment or (b) indicate whether they would share each piece of information. In the context of truth assessments, for each person, a *hit rate* can be calculated, how often they judge true information as true, as well as a *false alarm rate*, how often they judge false information as true. Discrimination sensitivity (d') is calculated as the difference between the standardized (Z -transformed) hit rate and false alarm rate—people with higher hit rates and lower false alarm rates are better able to discriminate true from false information when assessing veracity.

Response bias (c') is calculated as the negative average of the standardized hit and false alarm rates, indicating a person's general tendency to endorse information as true—regardless of its veracity. A higher d' score signals better discrimination; a higher c' score reflects a more conservative threshold for judging information as true.

In the context of sharing, a similar procedure yields a discrimination sensitivity score (d'), reflecting how much people differentiate sharing of true versus false information (i.e., how much their sharing diets include true information and do not include false information), and a response bias score (c'), which reflects a general tendency to share information—regardless of its veracity. Note that people share information for various reasons beyond accuracy (Pennycook et al., 2021); a lower sharing d' could therefore reflect either that people share false information, failing to discriminate the veracity of information, and/or that they simply do not share much true information, even though they may be good at detecting accuracy. Given the ambiguity in interpreting sharing intentions, it is important to measure both sharing and accuracy assessments, which we do in Study 2.

To examine the role of partisan bias, discrimination sensitivity and response thresholds can be calculated separately for politically congruent and incongruent information (Batailler et al., 2022; Gawronski et al., 2023). For example, people may exhibit a stronger partisan bias, as reflected by a lower response threshold for information that favors their political ingroup (e.g., Democrats evaluating anti-Trump information) versus the political outgroup (e.g., Democrats evaluating pro-Trump information). This means a greater propensity to share identity-congruent (vs. incongruent) information, regardless of information veracity.

Analyzing discrimination sensitivity and response thresholds separately for identity-congruent and incongruent information helps clarify the nature of older adults' misinformation sharing. To the extent that older adults share misinformation due to reduced analytical discrimination of true from false information, discrimination sensitivity should decrease with age. Further, older adults should have lower scores on measures of analytical thinking (e.g., need for cognition [NFC], cognitive reflection, faith in intuition; Bruine de Bruin et al., 2015; Cacioppo et al., 1996; Cacioppo & Petty, 1982), and these lower scores should statistically mediate the effect of age in predicting lower discrimination. To the extent that greater misinformation sharing is due to increased partisan bias, the inclination to share identity-congruent versus incongruent information (i.e., the difference between response thresholds for congruent and incongruent information) should increase with age. Although studies have investigated how age influences news' accuracy judgments (Allcott & Gentzkow, 2017; Pehlivanoglu et al., 2022; Sultan et al., 2024), none has systematically investigated why age shapes misinformation sharing—a critical gap addressed across our two studies.

The Present Studies

We conducted two studies to test whether age-related increases in misinformation sharing are associated with diminished differentiation of true and false information, stronger partisan bias, or both. In Study 1, conducted shortly after the 2022 U.S. midterm and Brazilian presidential elections, participants in both countries evaluated whether they would share true or false political news headlines that were congruent or incongruent with their political identities. Using SDT, we measured participants' discrimination sensitivity and response thresholds, including for both politically

congruent and incongruent information. We also measured analytical thinking. Study 2, conducted in the United States, extended this approach by including both sharing intentions and accuracy judgments. Study 2 also experimentally manipulated intuitive versus analytical thinking styles. Together, these studies examined whether age predicts weaker discrimination, stronger partisan bias, or differential response to thinking style prompts.

Pilot Study

In the weeks approaching the 2022 U.S. midterm elections and Brazil's presidential runoff, we searched for false and true news items for inclusion in a study to be conducted near the elections (Study 1). False items were sampled from fact-checking websites, whereas true news items were sampled from mainstream news media. Because Study 1 was conducted around the 2022 elections in both countries, headlines were selected for being related to elections to heighten relevance. For each country, we ended up with a total of 12 election-related news. News items were actual headlines that circulated around elections, although they were not systematically sampled from the population of news items available at the time.

We conducted a pretest in the U.S. sample where we asked 99 Prolific panelists which group each of the news items favored (1 = *definitely Democrats*, 5 = *definitely Republicans*). We wanted to confirm that participants would, on average, rate items selected by favoring Republicans as indeed favoring Republicans and items selected by favoring Democrats as favoring Democrats. We also measured familiarity with the news item (1 = *not familiar at all*, 5 = *very familiar*). Supplemental Material A shows the item-specific ratings (Supplemental Table S1). No pretest was conducted in Brazil due to operational challenges of attaining both a pretest and a study sample.

The results confirmed that participants rated items selected by favoring Republicans (vs. Democrats) as favoring Republicans more than Democrats ($M_{\text{Republican}} = 3.98$, $SD = 0.58$; $M_{\text{Democrat}} = 2.43$, $SD = 0.66$; $p < .001$). Further, although participants perceived true news items as more familiar than false news items, $M_{\text{true}} = 1.74$, $SD = 0.05$; $M_{\text{false}} = 1.41$, $SD = 0.04$, $t(98) = 8.60$, $p < .001$, across items, participants reported relatively low levels of familiarity ($M = 1.57$, $SD = 0.40$), with similar ratings for news favoring Republicans versus Democrats ($M_{\text{Republican}} = 1.59$, $SD = 0.42$; $M_{\text{Democrat}} = 1.56$, $SD = 0.50$, $p = .394$).

Transparency and Openness

For both studies, we report sample size, data exclusions, manipulations, and measures. Data, analysis code, and materials are available at https://osf.io/uwapy/?view_only=d877cc9697ce425590d9c0ab5ed14ee6. Data were analyzed using Stata 18, and all power analyses were conducted using the GPower software (Faul et al., 2007). This research received Institutional Review Board approval from the University of Colorado Boulder.

Study 1

In November 2022, shortly after that year's U.S. midterm elections and Brazil's presidential runoff, we asked respondents in both countries if they would share various election-related news that varied in veracity (true or false) and political slant (favoring Democrats or Republicans in the United States or liberals or conservatives in Brazil). We examined if

age predicted sharing discernment (d' score), the threshold for sharing news (c' score), and their political identity congruence. To examine whether analytical thinking predicted information sharing, we included measures of NFC (Cacioppo & Petty, 1982; Norris et al., 1998), the Cognitive Reflection Test (CRT; Frederick, 2005), and faith in intuition (Norris et al., 1998). We also measured education, which might be associated with political knowledge.

We conducted the study in Brazil and the United States, two democracies with, at the time, recent controversial presidential elections. It has been suggested that the United States's bipartisan system contributes to stronger partisanship (Finkel et al., 2020). By examining sharing in the United States and Brazil, a country with among the most fragmented party systems globally, our study examines whether partisan dynamics emerge across diverse political contexts (Flores et al., 2022).

Method

Participants

Power calculations suggested that sample sizes of 620 in each country would afford 80% power to detect within-country effect sizes of $d = .20$ in a one-tailed test using an α level of .05. Anticipating exclusions, we recruited 754 participants in the United States (Mechanical Turk) and 852 participants in Brazil (Netquest) between 22 and 29 November 2022. We excluded 22 (United States) and 155 (Brazil) participants who did not provide their age, identified as independent (center), or did not complete the political identification measure. The final sample comprised 732 U.S. and 697 Brazil participants. Importantly, all age groups are represented enough to afford adequate power to detect effects (for descriptive statistics and age distribution, see Supplemental Table S2 and Supplemental Figure S1).

Procedure

Participants read 12 election-related randomly ordered news headlines, distributed in a 2 (true vs. false) $\times$ 2 (favors Democrat/liberal vs. Republican/conservative identities) within-subjects design. For each headline, participants indicated their intention to share it: "How likely would you be to share this news in your social media?" (1 = *extremely unlikely*, 4 = *neutral*, 7 = *extremely likely*). For the SDT analysis, which uses dichotomous measures (Batailler et al., 2022; Gawronski et al., 2023; Pfänder & Altay, 2025), we categorized ratings as "yes" if higher than 4 and "no" if 4 or lower. We included 4 in the "no" category based on the conservative assumption that most people who report neutrality would rather not share the headline, given that most people rarely or never share content on social media (McClain, 2021). Indeed, no headline was shared by more than 50% of the participants (for news-specific sharing percentages, see Supplemental Material B, Table S3).

Next, participants completed a six-item CRT (Frederick, 2005; Thomson & Oppenheimer, 2016), which measures people's tendency to override their intuitions and think analytically. They also completed a five-item NFC scale (Cacioppo & Petty, 1982; Norris et al., 1998), which measures preferences for analytical thinking (Cronbach $\alpha_{\text{U.S.}} = .88$, $\alpha_{\text{BR}} = .87$), and a five-item Faith in Intuition scale measuring preferences for intuitive, holistic thinking (Cronbach $\alpha_{\text{U.S.}} = .89$, $\alpha_{\text{BR}} = .84$; see Norris et al., 1998). These measures were moderately correlated ($r_{\text{CRT-NFC}} = 0.13$, $r_{\text{CRT-Intuitive}} = -0.24$, $r_{\text{NFC-Intuitive}} = -0.10$, all $ps < .001$; see Pennycook et al., 2016).

As our measure of political identification, in each country, we used the most common criterion people use to define their political identities. In the United States, people's political identification is primarily as Democrat or Republican. We used an approach based on responses to the American National Election Studies questions to group people (including leaners) as Democrat or Republican (Flores et al., 2022; Westfall et al., 2015), excluding those who did not identify with or lean toward Democrats or Republicans.

In Brazil, people's political identification is primarily as liberal or conservative. Although Brazil is a highly polarized country (Furst et al., 2024; Ramos et al., 2025), identification with a political party is not a meaningful basis of political identification, in part because there are more than 30 parties (Ramos et al., 2025; Samuels & Zucco, 2014). Thus, we followed previous studies and used political ideology as our measure of political affiliation (1 = *very left-wing*, 4 = *center*, 7 = *very right-wing*; Ramos et al., 2020, 2025), excluding those who identified as center. Importantly, research shows that Brazilians not only identify themselves along the left-right continuum but also respond to source cues that share a political orientation (e.g., left- and right-wing politicians; Flores et al., 2022; Ramos et al., 2025). A recent cross-country study that included data from Brazil and the United States relied on a similar approach and found similar results for both countries (Flores et al., 2022), even though the criteria of political identity differed across these countries.

Last, participants reported gender, age, education, income, and race. Participants were thoroughly debriefed.

Data Treatment and Analysis Plan

Hit Rates and False Alarm Rates. We calculated hit rates (H) as the proportion of true headlines participants would share, and false alarm rates (FA) as the proportion of false headlines participants would share. We calculated each rate separately for items that were congruent and incongruent with participants' political affiliation. Following Macmillan and Creelman (2004), we converted proportions of 0 and 1 to $1/(2N)$ and $1 - 1/(2N)$, respectively, where N equals the number of trials on which the proportion is based (three headlines per condition in each country).

Discrimination Sensitivity and Response Threshold. Using SDT, we calculated discrimination sensitivity, $d' = z(H) - z(FA)$, and response threshold, $c' = -0.5 \times [z(H) + z(FA)]$, which include transformations of H and FA to a quantile function for a z distribution, with a proportion of 0.5 converted to a z -score of 0 (reflecting chance). The two indices were not correlated ($r = -.015$, $p = .560$). We computed d' and c' separately for congruent and incongruent news.

Regression Analyses. We separately predicted d' and c' in multiple regression models including age (standardized within-country), headline political identity congruence ($-.5$ = incongruent, $.5$ = congruent), a country indicator ($-.5$ = Brazil, $.5$ = United States), and all the two-way and three-way interactions between these three variables. We also included controls for household income (standardized within-country) and years of education (standardized within-country). Older people tend to have more income than younger adults, and education is commonly associated with age and negatively predicts misinformation belief (Delmastro & Paciello, 2022). In our data set, both were modestly correlated with age ($r_{\text{Age} \times \text{Income}} = 0.11$, $r_{\text{Age} \times \text{Education}} = 0.09$; for correlation matrix, see Table 1). Excluding the controls did not change the results (Supplemental Material B,

Table 1
Study 1: Correlation Between Key Measures

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Age	—											
2. Political conservatism	.115*	—										
3. Household income	.105*	.052	—									
4. Years of education	.094*	-.072*	.343*	—								
5. CRT	-.06*	-.039	.093*	.146*	—							
6. NFC	.078*	-.031	.092*	.082*	.129*	—						
7. Faith in intuition	-.092*	.119*	-.025	-.093*	-.236*	-.098*	—					
8. Party importance	.051	-.034	.02	.074*	.117*	.081*	-.024	—				
9. Discrimination sensitivity (d')	.049	-.112*	.013	.068*	.013	.027	.047	.047	—			
10. Response threshold (c')	-.013	-.210*	.038	.029	.110*	-.050	-.116*	-.254*	-.015	—		
11. Partisan discrimination (d' congruent– d' incongruent)	-.039	-.094*	-.014	-.004	.013	.003	-.034	.080*	-.177*	-.017	—	
12. Partisan bias (c' congruent– c' incongruent)	.069*	.166*	.015	-.015	.037	.142*	-.033	.232*	-.01	-.382*	.115*	—

Note. Samples include participants from Brazil and the United States. We report mean and standard deviations of all measures combined across countries, except household income, which is not easily comparable across countries. For country-specific descriptive statistics, see Supplemental Table S2. CRT = Cognitive Reflection Test; NFC = need for cognition; cong = congruent; incong = incongruent.
* $p < .05$.

Table S4). All models clustered standard errors at the individual level (see Table 2).

We also conducted additional analyses to examine if the significance remains unchanged across different specifications. First, our theoretical background and the main models are based on the participants' political identity. However, given that political ideology does not always align with people's political identities (e.g., Mason, 2015), we explored the inclusion of ideology as a control variable (Supplemental Material B, Table S5). In our data set, partisanship and ideology are highly correlated for the U.S. sample ($r = 0.83$, $p < .001$; note that we cannot calculate this correlation in the Brazilian sample, given that Brazilians use ideology as political identification). Although including ideology may raise collinearity issues, the effects were largely unchanged.

Second, we also analyzed the data using a different analytical approach. Specifically, we predicted the continuous 7-point measure of sharing intentions as a function of news veracity, identity congruence, and age, as well as all the two-way and three-way interactions and controls for income and education. We also clustered the standard errors at the levels of both the individual and news item. The key results were similar and reported in Supplemental Material B (also see Supplemental Table S6).

Exploratory Mediation Analyses. We explored whether measures of analytical thinking mediated the effects of age in predicting information sharing (Baron & Kenny, 1986). The main findings are summarized in the article, with the full step-by-step analysis provided in Supplemental Material B (also see Supplemental Tables S7–S9). We acknowledge that mediation analyses should be interpreted with caution, especially in correlational designs (see Bullock & Green, 2021; Fiedler et al., 2011; Zhao et al., 2010).

Additional Exploratory Analyses. We conducted additional exploratory analyses to inform various research questions. For example, a key finding in the misinformation literature is that Republicans and conservatives are more likely to believe in misinformation (Allcott & Gentzkow, 2017; Mosleh et al., 2024; Pennycook et al., 2020; Pennycook & Rand, 2019) and to share it (Guess et al., 2019) than Democrats and liberals. We thus investigated potential asymmetries in age-related effects on d' and c' based on the participant's political identity as a Democrat/liberal

versus Republican/conservative. We added a factor indicating the participant's political identity ($-.5$ = Democrat/liberal vs. $.5$ = Republican/conservative), as well as all its interactions with age, congruence, and country (Supplemental Material B, Table S10).

Additionally, given research showing that misinformation is shared primarily by strong partisans (Osmundsen et al., 2021), we explored whether partisan strength moderates age-related effects on d' and c' . From the same 7-point measure used to identify identity congruence (i.e., partisanship in the United States and ideology in Brazil), we created a measure of partisan strength centered on moderate strength (lean toward party/ideology = -0.5 , clear identification with party/ideology = 0 , strong identification with party/ideology = 0.5). We included this measure in the models predicting discrimination sensitivity (d') and sharing threshold (c'), along with all the two-way, three-way, and four-way interactions with age, congruence, and country (Supplemental Material B, Table S11).

Results

Age and Discrimination Sensitivity (d')

Age was not associated with lower discrimination sensitivity (d' ; see Table 2, Model 1, and Figure 1). Instead, d' increased with age ($b = 0.02$, $SE = 0.01$, 95% CI $[0.00, 0.04]$, $p = .049$). If anything, then, older adults were more, not less, discriminating in deciding to share true versus false headlines, consistent with the conclusions of a recent meta-analysis (Sultan et al., 2024).

There was a significant effect of identity congruence on discrimination sensitivity ($b = -0.11$, $SE = 0.02$, 95% CI $[-0.16, -0.07]$, $p < .001$). People were more discerning of true and false information when deciding to share information that was incongruent rather than congruent with their political identities (Figure 1). However, age did not significantly interact with congruence ($b = 0.04$, $SE = 0.02$, 95% CI $[-0.003, 0.08]$, $p = .069$), which fails to provide evidence that older adults more than younger adults differentially discriminate information based on congruence with their political identity. These effects remain unchanged whether we include political ideology as a control variable or exclude all controls (see Supplemental Material B, Tables S4 and S5).

Table 2

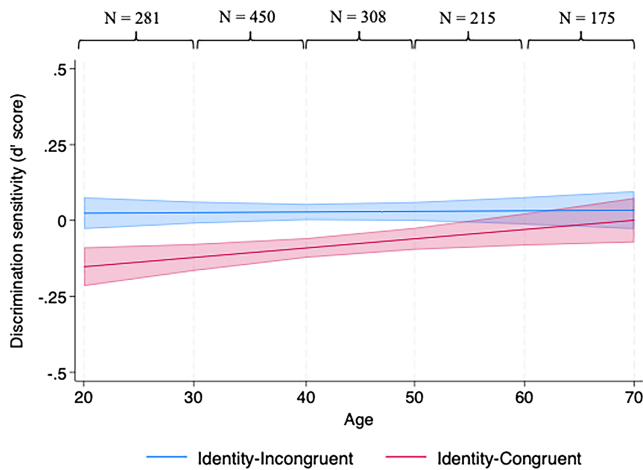
The Effects of Age, Partisan Bias, and Country on d' and c' Scores for Sharing Intentions (Study 1)

Variable	Discrimination sensitivity (d')			Response threshold (c')		
	Model 1			Model 2		
	Coefficient	SE	p	Coefficient	SE	p
Age	0.021	0.011	.049	−0.015	0.014	.282
Congruence	−0.113	0.022	<.001	−0.404	0.017	<.001
Brazil_U.S.	−0.029	0.021	.176	0.359	0.027	<.001
Age × Congruence	0.039	0.021	.069	−0.045	0.017	.010
Age × Brazil_U.S.	0.000	0.022	.987	0.021	0.027	.436
Congruence × Brazil_U.S.	0.018	0.043	.679	0.336	0.033	<.001
Age × Congruence × Brazil_U.S.	−0.037	0.042	.385	−0.031	0.035	.365
Household income	−0.007	0.011	.482	0.016	0.014	.279
Years of education	0.030	0.011	.007	0.024	0.015	.108
Constant	−0.028	0.011	.009	0.456	0.013	.000

Note. Age is standardized within-country. Congruence and Brazil_U.S. are contrast coded. SE = standard error.

Figure 1

Study 1: Predicted Effects of Age and Political Identity Congruence on Discrimination Sensitivity (d' Score)



Note. Discrimination sensitivity (d') is a score that reflects how well participants differentiate true from false information when deciding what to share. Higher d' scores reflect greater truth discrimination in respondents' sharing diets. Coefficients were predicted from the regression Model 1 in Table 2, which uses within-country standardized age. For reading ease, the labels in the graph display raw age, which was determined by cross-tabulating standardized and raw ages in the sample, regardless of country. Shaded areas reflect 95% confidence intervals. See the online article for the color version of this figure.

Age and Response Threshold (c')

We next examined the response threshold for sharing information (c' score; see Figure 2 and Table 1, Model 2). Age did not predict lower response thresholds ($b = -0.01$, $SE = 0.01$, 95% CI $[-0.04, 0.01]$, $p = .282$), providing no evidence that older adults are simply more likely to share information. A negative effect of identity congruence indicated that people had a higher threshold for sharing identity-incongruent than identity-congruent information ($b = -0.40$, $SE = 0.02$, 95% CI $[-0.44, -0.37]$, $p < .001$). People displayed a partisan bias, in other words, being less likely to share incongruent than congruent information.

Importantly, there was an interaction between age and political identity congruence ($b = -0.04$, $SE = 0.02$, 95% CI $[-0.08, -0.01]$, $p = .010$; see Figure 2 and Table 1, Model 2). Spotlight analyses examining the effect of congruence at 1 standard deviation below and above the mean age demonstrated a larger effect of congruence for older (55 years; $b = -0.44$, $SE = 0.02$, 95% CI $[-0.49, -0.39]$, $p < .001$) than younger individuals (29 years; $b = -0.35$, $SE = 0.02$, 95% CI $[-0.40, -0.31]$, $p < .001$). Unpacking the interaction differently, age predicted lower thresholds for sharing identity-congruent information ($b = -0.04$, $SE = 0.02$, 95% CI $[-0.07, -0.00]$, $p = .050$), but age did not significantly predict sharing thresholds for identity-incongruent information ($b = 0.01$, $SE = 0.01$, 95% CI $[-0.02, 0.03]$, $p = .559$). Older (vs. younger) adults were thus more partisan in their intentions to share congruent over incongruent information. These effects remain unchanged whether we include political ideology as a control variable or exclude all controls (see Supplemental Material B, Tables S4 and S5).

Exploratory Mediation Analyses

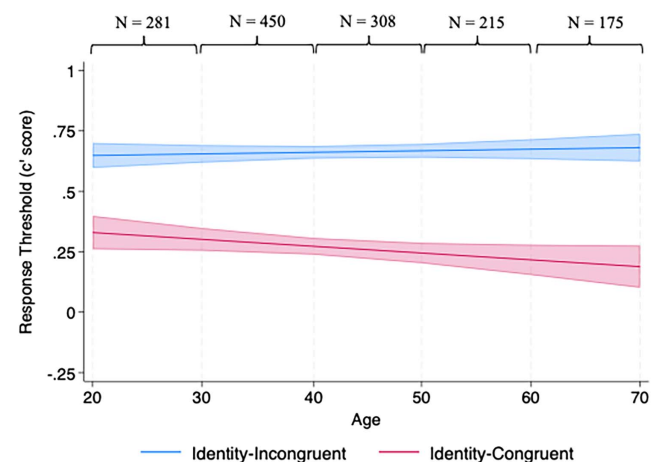
Age did not predict lower discrimination of sharing true versus false information (d'), differential discrimination of congruent versus incongruent information, nor overall response thresholds (c'). Instead, age predicted an increased partisan bias, where age and congruence interactively predicted sharing thresholds (c'). Thus, we explore whether measures of analytical thinking (NFC, CRT, and faith in intuition) mediate this interaction effect. For a detailed analysis, see Supplemental Material B.

NFC significantly interacted with congruence in predicting response threshold ($b = -0.09$, $SE = 0.02$, 95% CI $[-0.13, -0.06]$, $p < .001$; see Supplemental Material B, Table S7). People with higher NFC, like older adults in our sample, had a relatively lower threshold for sharing ideologically congruent versus incongruent information. However, the relevant Age $\times$ Congruence interaction remained significant ($b = -0.04$, $SE = 0.02$, 95% CI $[-0.07, -0.004]$, $p = .029$) and was not significantly reduced ($Z = -0.31$, $p = .753$). People's NFC, like age, was thus associated with a greater partisan bias in thresholds for sharing information, but there was no evidence that NFC mediates the effect of age interacting with congruence in predicting sharing thresholds.

CRT did not significantly interact with congruence in predicting response threshold ($b = -0.01$, $SE = 0.02$, 95% CI $[-0.04, 0.02]$, $p = .408$; see Supplemental Table S8), and the Age $\times$ Congruence interaction remained significant and was not significantly reduced when CRT and relevant interactions were included in the model ($b = -0.05$, $SE = 0.02$, 95% CI $[-0.08, -0.01]$, $p = .009$; $Z = -0.31$, $p = .757$). Similarly, faith in intuition did not significantly interact with congruence to predict response threshold ($b = 0.01$, $SE = 0.02$,

Figure 2

The Effects of Age and Identity Congruence on Response Bias (c' Score)



Note. Response bias (c') reflects the tendency to share information. Higher c' scores reflect a higher threshold to share the headline, indicating that participants are generally more hesitant to share. Coefficients were predicted from the regression Model 2 in Table 2, which uses within-country standardized age. For reading ease, the labels in the graph display raw age, which was determined by cross-tabulating standardized and raw ages in the sample, regardless of country. Shaded areas reflect 95% confidence intervals. See the online article for the color version of this figure.

95% CI $[-0.02, 0.04]$, $p = .538$; see [Supplemental Table S9](#)), and the Age $\times$ Congruence interaction remained significant and was not significantly reduced when faith in intuition and relevant interactions were included in the model ($b = -0.04$, $SE = 0.02$, 95% CI $[-0.08, -0.01]$, $p = .015$; $Z = -0.11$, $p = .911$). Unlike age, then, CRT and faith in intuition were not significantly associated with partisan bias in sharing information.

Comparison of United States and Brazil

The key results were consistent in the United States and Brazil. There were no significant interactions between country and age in predicting discrimination sensitivity ($b = -0.00$, $SE = 0.02$, 95% CI $[-0.04, 0.04]$, $p = .987$) or response threshold ($b = 0.02$, $SE = 0.03$, 95% CI $[-0.03, 0.07]$, $p = .436$). Nor were there three-way interactions between country, age, and identity congruence for either outcome (d' : $b = -0.04$, $SE = 0.04$, 95% CI $[-0.12, 0.05]$, $p = .385$; c' : $b = -0.03$, $SE = 0.03$, 95% CI $[-0.10, 0.04]$, $p = .365$). The only differences were a higher response threshold ($b = 0.36$, $SE = 0.03$, 95% CI $[0.31, 0.41]$, $p < .001$) and a weaker effect of congruence in the United States than Brazil ($b = 0.34$, $SE = 0.03$, 95% CI $[0.27, 0.40]$, $p < .001$), but these effects are not related to age, which is the focus of this article.

Political Identity

Republicans in the United States and conservatives in Brazil were less discriminating of true and false information than Democrats and liberals ($b = -0.05$, $SE = 0.02$, 95% CI $[-0.09, -0.01]$, $p = .024$). However, political identity did not significantly interact with any effects of age in predicting discrimination scores.

Considering the c' score, there was a main effect such that Republicans/conservatives had lower sharing thresholds than Democrats/liberals ($b = -0.13$, $SE = 0.03$, 95% CI $[-0.19, -0.08]$, $p < .001$). A significant three-way interaction between age, congruence, and identity emerged ($b = -0.08$, $SE = 0.04$, 95% CI $[-0.15, -0.01]$, $p = .030$), indicating the key effect of age increasing partisan bias was stronger among Republicans/conservatives than among Democrats/liberals.

Strength of Political Identity

For the d' score, the three-way interaction between age, congruence, and partisan strength was not significant ($b = -0.08$, $SE = 0.05$, 95% CI $[-0.17, 0.01]$, $p = .096$). However, this interaction significantly predicted response thresholds c' ($b = -0.10$, $SE = 0.04$, 95% CI $[-0.17, -0.02]$, $p = .010$). The Age $\times$ Congruence interaction was significantly larger for strong partisans ($b = -0.11$, $SE = 0.03$, 95% CI $[-0.17, -0.04]$, $p = .001$) than for leaning partisans ($b = 0.003$, $SE = 0.02$, 95% CI $[-0.04, 0.05]$, $p = .879$).

Discussion

Study 1 found that age increases partisan bias in sharing information. Although older adults were not more likely to share information as younger adults (i.e., there was no effect of age on response thresholds), older adults, especially those who report being strong partisans, were significantly more likely than younger adults to sharing identity-congruent information more than incongruent information.

The study also found evidence that higher analytical thinking (measured with NFC) predicted a stronger partisan bias. We did not, however, find that analytical thinking mediated or moderated the age-related effects on partisan bias. Further, the study demonstrated that having a Republican/conservative (vs. Democrat/liberal) identity was associated with lower discrimination sensitivity and response thresholds and increased age-related partisan bias. Partisanship strength also reduced response thresholds and increased partisan bias, especially among older adults.

Age did not reduce sharing discernment of true versus false information. Not only did cognitive reflection and preferences for analytical and intuitive thinking fail to significantly predict sharing discernment, the effect of age on discernment was largely unchanged when these analytic measures were included in the model. These results failed to provide support for the information discernment and analytical thinking explanation of why older adults share more misinformation.

Study 1 provides evidence that older adults in Brazil and the United States display stronger partisan bias in misinformation sharing, yet the study has several limitations. First, by only measuring sharing intentions, it remains unclear whether similar results would be observed for accuracy judgments. Previous research suggests that decisions to share news items do not necessarily correspond with accuracy assessments (e.g., Pennycook et al., 2021; Pennycook & Rand, 2021). Thus, it could be the case that older people were more biased in sharing news because they were more biased in assessing the accuracy of news items. Alternatively, people may share information they believe to be false, deliberately spreading fake news, or they may not share what they believe to be true. Measuring both intentions to share and assessment of information accuracy would allow us to address these interpretations. Second, in Study 1, sharing intentions was measured on a 7-point scale, which we converted to dichotomous measure of sharing (yes/no) using the conservative assumption that those who were neutral would likely not be willing to share the news. A binary choice would avoid such assumptions. Third, Study 1 was conducted right after elections in the United States and Brazil, in periods with heightened political identity salience. It is an open question whether similar effects would generalize to other time periods. Study 2 addresses these issues.

Study 2

We conducted Study 2 with two key goals, in addition to replicating the findings from Study 1. First, we measured both people's intentions to share news headlines, as in Study 1, as well as their assessments of the accuracy of news headlines. Previous research suggested that factors shaping decisions to share fake news are distinct from assessments of veracity, suggesting that people may be inclined to share news they would have assessed as fake—and not to share news they assess as true (Pennycook et al., 2021; Pennycook & Rand, 2021). Measuring both sharing and accuracy assessments can inform understanding of the nature of information sharing. If age and partisanship predict intentions to share news but not assessment of accuracy, that implicates processes related to socio-political communication considerations, not information evaluation. Alternatively, if age and partisan congruence predict both intentions to share news and assessments of accuracy, that implicates processes related to partisan information evaluation, not necessarily grounded in communication considerations.

Second, Study 2 included an exploratory experimental manipulation prompting people to think either intuitively or analytically (Bago et al., 2020; Gärtner et al., 2022; Gawronski et al., 2023). Study 1 did not find evidence that widely used measures of analytical thinking mediated age-related increases in partisan information sharing. However, it is possible that temporarily induced thinking styles, distinct from stable individual differences in thinking styles, influence such biases and that older people are more influenced by transient factors that shape thinking style. We therefore tested whether participants prompted to engage in analytical thinking would exhibit reduced partisan bias in assessing the accuracy of and willingness to share politically congruent headlines, relative to those prompted to engage in intuitive thinking, and whether this effect would be stronger among older adults. We also explored whether analytical thinking prompts would increase overall accuracy discrimination, reducing the tendency to more stringently discriminate identity-incongruent headlines, and whether this effect would be larger among older adults.

Method

Participants

Power calculations suggested a sample size of 620 participants to achieve .80 power to detect a significant small effect ($d = .20$) in a one-tailed test using an α level of .05. In May 2023, we recruited 1,120 participants in the United States through Mechanical Turk to increase power to detect effects considering potential exclusions (for descriptive statistics and age distribution, see Supplemental Table S12 and Supplemental Material C, Figure S2). We excluded 136 participants from the analyses because they failed to assess the news headlines, had missing age, or did not identify as Democrat or Republican. The final sample comprised 984 participants.

Procedure

After providing consent, participants were randomly assigned to one of three conditions: control, analytical thinking prime, or intuitive thinking prime. The procedure in the control condition was nearly identical to the main study, with key exceptions noted below. Based on previous research (Gärtner et al., 2022), participants in the two treatment conditions were prompted to think in an analytical or intuitive way. Specifically, before proceeding to the news evaluation task, participants in the analytical condition were prompted with the following (*intuitive reasoning condition in italics*):

People have different preferences when it comes to thinking styles. More often than not people prefer making decisions by using reason and relying on their heads, carefully analyzing the situations [*by using their intuitions and gut feelings, relying on their hearts*]. Many people believe that the head [*heart*] is the part of our body that is most connected with good decision-making. When we think with our heads [*feel with our hearts*], we make satisfying decisions. In your own words, please explain briefly why you think analytical reasoning [*intuition*] is often the best thinking style.

In the control condition, participants proceeded directly to the news evaluation task.

The news evaluation task was similar to Study 1, with a key difference in measurement. For each news item, participants reported whether they would share the news on their social media (1 = yes,

0 = no) and whether the claim in the news headline was, to the best of their knowledge, true or false (1 = true, 0 = false). Participants indicated whether they would share each of the items in one block and indicated whether each item was accurate in another block. The order of sharing and accuracy blocks was counterbalanced to avoid contamination, and our analysis considered only the outcome which they have seen first (for news-specific accuracy and sharing percentages, see Supplemental Material C, Supplemental Table S13). After this task, the participants completed the same measures of Study 1.

Data Treatment and Analysis Plan

Discrimination Sensitivity and Response Threshold. Following the same procedures as in Study 1, we calculated discrimination sensitivity (d' score) and response threshold (c' score) indices for sharing or accuracy, depending on which item they answered first, as well as separate scores for identity-congruent and incongruent news. We only analyzed the type of measure participants completed first to minimize pressures for consistency in reporting sharing and accuracy assessments. Participants who were asked about accuracy first, for example, might have reported sharing intentions with accuracy salient in mind, or to self-present as only sharing accurate information (Capraro & Celadin, 2023; Pennycook et al., 2020, 2021; Pennycook & Rand, 2022). Discrimination sensitivity (d') and response thresholds (c') were not correlated for sharing ($r = -0.063$, $p = .162$) or for accuracy judgments ($r = 0.058$, $p = .197$; see Table 3).

Regression Analyses. We regressed discrimination sensitivity (d') and response threshold (c') index each on age (standardized), a contrast-coded variable indicating whether the index refers to the identity-congruent or identity-incongruent set of news ($-.5 =$ incongruent, $.5 =$ congruent), a contrast-coded variable indicating the index's type of measure ($-.5 =$ sharing, $.5 =$ accuracy), all the two-way and three-way interactions between these three factors, as well as controls for household income (standardized) and years of education (standardized). We also included in the model two contrast codes for conditions (intuitive_analytical = $-.5$ if intuitive condition, $= 0$ if control condition, $= .5$ if analytical condition; treatment_control = $-1/3$ if intuitive condition, $= 2/3$ if control condition, $= -1/3$ if analytical condition), as well as their interactions with the preceding variables (see Table 4). We clustered the standard errors at the individual level. Excluding the controls did not change the results (Supplemental Material C, Table S14).

As in Study 1, despite the high correlation between ideology and partisanship ($r = 0.73$, $p < .001$), including ideology as a control variable did not change the results (Supplemental Material C, Table S15). Likewise, a non-SDT model that predicts belief or sharing likelihood from age, political identity congruence, news accuracy, type of outcome (sharing vs. accuracy judgment), and the experimental manipulations, as well as controls for income and education, yields similar results (Supplemental Material C, Table S16).

Exploratory Mediation Analyses. We also conducted exploratory mediation analyses, as in Study 1. The main findings are summarized below, with the step-by-step analysis provided in Supplemental Materials C (also see Supplemental Tables S17 and S18). Again, we acknowledge that mediation analyses should be

Table 3
Study 2: Correlation Between Key Measures

Variable	<i>M (SD)</i>	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Age	43.12 (12.71)	—															
2. Political conservatism	0.35 (0.48)	.099*	—														
3. Household income	5.579 (3.295)	-.017	.037	—													
4. Years of education	15.63 (1.99)	.029	-.049	.348*	—												
5. CRT	0.66 (0.31)	.037	-.020	.032	.018	—											
6. NFC	5.87 (0.82)	.073*	-.020	.045	-.069*	.169*	—										
7. Faith in intuition	3.85 (1.46)	-.153*	.009	.021	.085*	-.364*	-.292*	—									
8. Party importance	2.93 (1.28)	.018	-.044	-.011	.168*	-.179*	-.023	.183*	—								
9. Sharing discrimination (<i>d'</i>)	0.01 (0.41)	.006	-.106*	-.072	-.019	-.005	-.071	-.017	.023	—							
10. Sharing threshold (<i>c'</i>)	0.48 (0.54)	.001	-.092*	.073	-.198*	.314*	.128*	-.296*	-.407*	-.063	—						
11. Sharing discrimination bias (<i>d'</i> cong- <i>d'</i> incong)	0.14 (0.74)	.021	-.028	.066	.009	-.007	.011	.058	.068	-.216*	-.061	—					
12. Sharing partisan bias (<i>c'</i> cong- <i>c'</i> incong)	0.25 (0.52)	.181*	.126*	.054	-.083	.081	.074	-.158*	.063	-.07	-.183*	.119*	—				
13. Accuracy discrimination (<i>d'</i>)	0.53 (0.55)	.049	-.011	.007	-.091*	.204*	.116*	-.243*	-.110*					—			
14. Accuracy threshold (<i>c'</i>)	0.11 (0.36)	.035	-.029	.104*	-.106*	.153*	.038	-.199*	-.276*				.058		—		
15. Accuracy discrimination bias (<i>d'</i> cong- <i>d'</i> incong)	0.13 (0.96)	-.009	-.265*	-.0134	-.031	-.012	.028	-.013	.059				-.102*	-.077		—	
16. Accuracy partisan bias (<i>c'</i> cong- <i>c'</i> incong)	0.37 (0.62)	.096*	-.197*	.008	-.036	.087	.132*	-.087	.065				-.016	-.056	.071		—

Note. Sample includes participants from the United States. Means and correlations involving indices (discrimination sensitivity, response thresholds, and the differences of these outcomes for congruent and incongruent headlines) related to sharing intentions only considered participants who reported their sharing intentions before providing accuracy judgments. Similarly, means and correlations of indices related to accuracy judgments only considered participants who provided accuracy judgments before reporting sharing intentions. Missing correlation coefficients reflect the between-subjects nature of the analysis; for a given participant, we only considered the type of outcome the participant answered first. CRT = Cognitive Reflection Test; NFC = need for cognition; cong = congruent; incong = incongruent.

* $p < .05$.

interpreted with caution, especially in correlational designs (see Bullock & Green, 2021; Fiedler et al., 2011; Zhao et al., 2010).

Additional Exploratory Analyses. We also investigated asymmetries in age-related effects on d' and c' based on the participant's political identity as a Democrat or Republican (Supplemental Material C, Table S19). Finally, we conducted an analysis analogous to that in Study 1 exploring whether partisan strength moderates age-related effects on d' and c' (Supplemental Material C, Table S20).

Results

Age and Discrimination Sensitivity (d')

As in Study 1, we again did not find support for the prediction of the analytical thinking account that age is associated with lower discrimination sensitivity d' . While Study 1 found a positive effect of age, there was a null effect in Study 2 ($b = 0.01$, $SE = 0.02$; 95% CI $[-0.02, 0.04]$, $p = .432$; see Figure 3 and Table 4, Model 1).

An effect of outcome type reflected that participants were more discriminating in assessing the accuracy of news items ($M = 0.48$, $SD = 0.75$) than in deciding whether to share news items ($M = 0.02$, $SD = 0.54$; $b = 0.52$, $SE = 0.03$, 95% CI $[0.46, 0.58]$, $p < .001$). This suggests that people share news they would otherwise categorize as

false or that they do not share news they would categorize as true. Importantly, however, age did not interact with sharing versus accuracy judgments ($b = 0.02$, $SE = 0.03$, 95% CI $[-0.04, 0.08]$, $p = .519$).

There was a significant main effect of identity congruence in predicting discrimination sensitivity ($b = -0.14$, $SE = 0.03$, 95% CI $[-0.20, -0.09]$, $p < .001$, Figure 3). People displayed greater discernment for news that was incongruent with their political identities ($M = 0.31$, $SD = 0.65$) than for information that was congruent with their identities ($M = 0.20$, $SD = 0.73$). However, age did not interact with identity congruence in predicting discrimination sensitivity ($b = -0.01$, $SE = 0.03$, 95% CI $[-0.06, 0.05]$, $p = .844$), nor did identity congruence interact with sharing decisions versus accuracy assessments ($b = 0.01$, $SE = 0.06$, 95% CI $[-0.10, 0.12]$, $p = .848$).

Age and Response Threshold (c')

We next examined response thresholds (c' score; see Figure 4 and Table 4, Model 2). As in Study 1, age was not associated with lower response thresholds ($b = 0.01$, $SE = 0.01$, 95% CI $[-0.01, 0.04]$, $p = .469$). Nor did age interact with sharing versus accuracy measures ($b = 0.01$, $SE = 0.03$, 95% CI $[-0.04, 0.07]$, $p = .606$). These results

Table 4

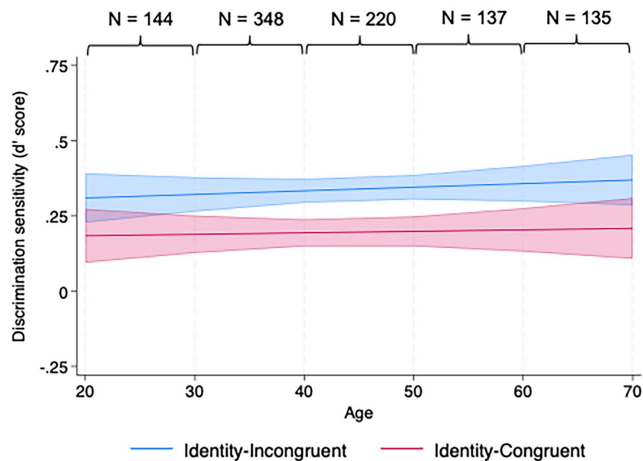
The Effects of Age, Partisan Bias, Measure Type, and Experimental Manipulations on d' and c' Scores (Study 2)

Variable	Discrimination sensitivity (d') Model 1			Response threshold (c') Model 2		
	Coefficient	SE	p	Coefficient	SE	p
Age	0.012	0.015	.432	0.010	0.014	.469
Congruence	-0.142	0.028	<.001	-0.314	0.018	<.001
Share_accuracy	0.517	0.031	<.001	-0.362	0.029	<.001
Intuitive_analytical	0.027	0.039	.484	0.039	0.036	.273
Treat_control	0.017	0.033	.599	0.018	0.031	.562
Age × Congruence	-0.006	0.028	.844	-0.073	0.019	<.001
Age × Share_Accuracy	0.020	0.030	.519	0.015	0.029	.606
Age × Intuitive_Analytical	-0.014	0.037	.697	0.069	0.036	.052
Age × Treat_Control	0.026	0.032	.420	-0.021	0.030	.477
Congruence × Share_Accuracy	0.011	0.056	.848	-0.114	0.036	.002
Congruence × Intuitive_Analytical	0.012	0.068	.857	0.108	0.046	.020
Congruence × Treat_Control	0.006	0.059	.925	-0.012	0.037	.744
Share_Accuracy × Intuitive_Analytical	0.153	0.078	.050	0.010	0.071	.888
Share_Accuracy × Treat_Control	0.017	0.065	.791	-0.038	0.062	.542
Age × Congruence × Share_Accuracy	0.019	0.056	.733	0.042	0.037	.265
Age × Congruence × Intuitive_Analytical	0.045	0.068	.506	0.055	0.049	.257
Age × Congruence × Treat_Control	0.072	0.073	.325	-0.048	0.071	.504
Age × Share_Accuracy × Intuitive_Analytical	0.151	0.136	.266	0.312	0.092	.001
Age × Share_Accuracy × Treat_Control	-0.119	0.059	.045	-0.008	0.037	.834
Congruence × Share_Accuracy × Intuitive_Analytical	0.114	0.065	.077	-0.017	0.060	.771
Congruence × Share_Accuracy × Treat_Control	-0.099	0.118	.402	-0.054	0.075	.469
Age × Congruence × Share_Accuracy × Intuitive_Analytical	0.040	0.137	.771	-0.002	0.097	.982
Age × Congruence × Share_Accuracy × Treat_Control	-0.012	0.119	.916	0.143	0.074	.055
Household income	-0.002	0.017	.903	0.077	0.015	<.001
Years of education	-0.028	0.017	.089	-0.102	0.016	<.001
Constant	0.270	0.016	<.001	0.295	0.014	.000

Note. Age, household income, and years of education are standardized. Congruence, Share_accuracy, Intuitive_analytical, and Treatment_control are contrast coded. *SE* = standard error.

Figure 3

Study 2: The Effects of Age and Identity Congruence on Discrimination Sensitivity (d' Score)



Note. Discrimination sensitivity (d') is a score that reflects how well participants differentiate true from false information when deciding what to share. Higher d' scores reflect greater discernment in respondents' decisions to share true news items and not share false news items. Coefficients were predicted from the regression Model 1 in Table 4, which uses within-country standardized age. For reading ease, the labels in the graph display raw age, which was determined by cross-tabulating standardized and raw ages in the sample. Shaded areas reflect 95% confidence intervals. See the online article for the color version of this figure.

suggest that older people are not simply more likely to share news items and judge them as accurate.

As in Study 1, people displayed a strong partisan bias in response thresholds, as reflected by a main effect of congruence ($b = -0.31$, $SE = 0.02$, 95% CI $[-0.35, -0.28]$, $p < .001$). The effect of congruence emerged for both sharing decisions ($b = -0.26$, $SE = 0.02$, 95% CI $[-0.30, -0.21]$, $p < .001$) and accuracy assessments ($b = -0.37$, $SE = 0.03$, 95% CI $[-0.43, -0.32]$, $p < .001$), with a significant interaction indicating the congruence effect was larger for assessing accuracy than for sharing decisions ($b = -0.11$, $SE = 0.04$, 95% CI $[-0.19, -0.04]$, $p = .002$). That is, people were more likely to judge as accurate and share information that was congruent rather than incongruent with their political identities. And people exhibited a larger partisan bias in assessing the accuracy of news than in sharing news.

Critically, replicating Study 1, partisan bias increased with age, as evidenced by a significant interaction between age and political identity congruence ($b = -0.07$, $SE = 0.02$, 95% CI $[-0.11, -0.04]$, $p < .001$; see Figure 4 and Table 4, Model 2). Spotlight analyses demonstrate that partisan bias was stronger for older adults (56 years; $b = -0.39$, $SE = 0.03$, 95% CI $[-0.44, -0.34]$, $p < .001$) than younger adults (30 years; $b = -0.24$, $SE = 0.02$, 95% CI $[-0.29, -0.19]$, $p < .001$). The interaction had a slightly different form than Study 1. Age significantly predicted a higher threshold for identity-incongruent news ($b = 0.05$, $SE = 0.02$, 95% CI $[0.02, 0.08]$, $p = .001$), but age did not predict a significantly lower threshold for identity-congruent news ($b = -0.02$, $SE = 0.02$, 95% CI $[-0.06,$

$0.01]$, $p = .229$); Study 1 found the reverse pattern. These results suggest that age is consistently associated with increased partisan bias in response thresholds, although the manifestation of this pattern, whether increasing thresholds for incongruent information or decreasing thresholds for congruent information, is less consistent.

There was no evidence that age-related partisan bias was significantly different for decisions about accuracy and sharing. Nor was the three-way interaction between age, congruence, and type of outcome not significant ($b = 0.04$, $SE = 0.04$, 95% CI $[-0.03, 0.11]$, $p = .265$).

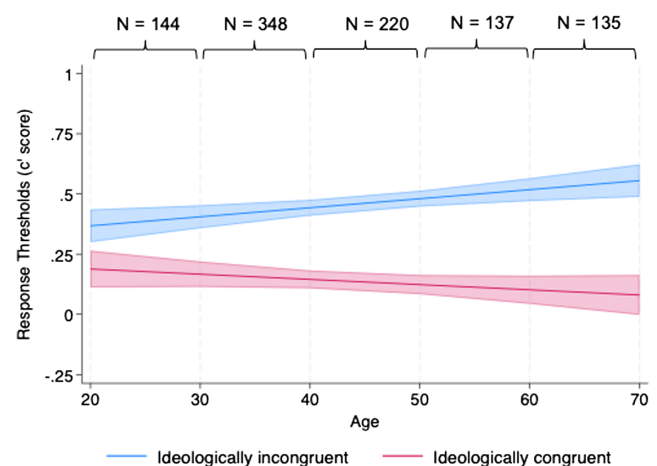
Experimentally Manipulated Thinking Styles

We first explored whether the manipulations affected measures of individual differences. There was no effect of condition on CRT and NFC scores, CRT: $F(2, 981) = 1.27$, $p = .280$; NFC: $F(2, 984) = 0.75$, $p = .473$, but intuitive thinking style was significantly higher in the intuitive condition ($M = 4.08$, $SD = 1.47$) than in the control condition, $M = 3.80$, $SD = 1.45$; $F(1, 665) = 6.16$, $p = .013$, and analytical thinking style condition, $M = 3.68$, $SD = 1.43$; $F(1, 629) = 12.28$, $p < .001$.

The experimental manipulations did not influence discrimination sensitivity ($b = 0.03$, $SE = 0.04$, 95% CI $[-0.05, 0.10]$, $p = .484$) or response threshold ($b = 0.04$, $SE = 0.03$, 95% CI $[-0.03, 0.11]$, $p = .273$; see Table 4). Furthermore, in predicting discrimination sensitivity, there was no two-way interaction between age and the manipulation nor a three-way interaction with congruence and type of outcome (see Table 4).

Figure 4

Study 2: The Effects of Age and Political Identity Congruence on Response Threshold (c' Score)



Note. Response bias (c') reflects the tendency to respond affirmatively (e.g., share or saying the news is true). Higher c' scores reflect a higher threshold to share the headline or to believe in the headline, indicating that participants are generally more hesitant to share or to say it is true. Coefficients were predicted from the regression Model 1 in Table 4, which uses within-country standardized age. For reading ease, the labels in the graph display raw age, which was determined by cross-tabulating standardized and raw ages in the sample. Shaded areas reflect 95% confidence intervals. See the online article for the color version of this figure.

There was a significant interaction between analytical versus intuitive conditions and type of outcome ($b = 0.15$, $SE = 0.08$, 95% CI [0.00, 0.31], $p = .050$). The analytical thinking condition increased people's discrimination sensitivity more for accuracy judgments ($b = 0.10$, $SE = 0.06$, 95% CI [-0.02, 0.23], $p = .096$) than sharing intentions ($b = -0.05$, $SE = 0.05$, 95% CI [-0.14, 0.04], $p = .294$). There were no other main effects or interactions involving experimentally manipulated thinking style on discrimination sensitivity.

Considering response threshold (c'), there was a significant two-way interaction between political identity congruence and the manipulation of intuitive or analytical thinking ($b = 0.11$, $SE = 0.05$, 95% CI [0.02, 0.20], $p = .020$). Participants in the analytical thinking condition displayed a weaker partisan bias ($b = -0.26$, $SE = 0.03$, 95% CI [-0.32, -0.19], $p < .001$) than those in the intuitive thinking condition ($b = -0.36$, $SE = 0.03$, 95% CI [-0.43, -0.30], $p < .001$). There was also a three-way interaction between congruence, the manipulation, and type of measure ($b = 0.31$, $SE = 0.09$, 95% CI [0.13, 0.49], $p = .001$). The interaction noted earlier, where people exhibited a weaker partisan bias in the analytical thinking condition compared with the intuitive thinking condition, was especially pronounced for accuracy judgments ($b = 0.26$, $SE = 0.08$, 95% CI [-0.12, 0.40], $p < .001$) compared with sharing intentions ($b = -0.05$, $SE = 0.06$, 95% CI [-0.16, 0.07], $p = .412$).

Experimentally manipulated thinking style did not interact with age and partisan congruence ($b = 0.06$, $SE = 0.05$, 95% CI [-0.04, 0.15], $p = .257$). This once again fails to provide evidence that age-related increases in partisan sharing and accuracy assessment are related to changes in analytical or intuitive thinking. If it did, then the age-related partisanship effect might have been smaller when participants had been primed to think analytically.

Exploratory Mediation Analyses

We explored potential mediators of the Age $\times$ Congruence interaction predicting response thresholds (c'). NFC significantly interacted with congruence in predicting response threshold ($b = -0.06$, $SE = 0.02$, 95% CI [-0.09, -0.02], $p = .001$; see Supplemental Table S17). People with higher NFC, like older adults, had a relatively lower threshold for sharing identity-congruent versus incongruent information. However, the relevant Age $\times$ Congruence interaction remained significant and was not significantly reduced ($b = -0.07$, $SE = 0.02$, 95% CI [-0.11, -0.04], $p < .001$; $Z = -0.04$, $p = .971$). People's NFC, like age, was associated with a greater partisan bias in thresholds for sharing information, but NFC did not mediate the effect of age interacting with congruence in predicting sharing thresholds.

Intuitive thinking also significantly interacted with congruence in predicting response threshold ($b = 0.06$, $SE = 0.02$, 95% CI [0.02, 0.09], $p = .001$; see Supplemental Table S18). People with a greater (vs. weaker) preference for intuitive thinking displayed a lower partisan bias, which is perhaps counter to intuition. However, the Age $\times$ Congruence interaction remained significant and was not significantly reduced ($b = -0.06$, $SE = 0.02$, 95% CI [-0.10, -0.03], $p = .001$; $Z = -0.38$, $p = .701$). Thus, we did not find evidence that NFC or a preference for intuitive thinking mediated the effect of age interacting with congruence in predicting sharing thresholds.

Political Identity

As in Study 1, Republicans were less discriminating of true and false information than Democrats and liberals, but this effect was not significant ($b = -0.06$, $SE = 0.03$, 95% CI [-0.13, 0.002], $p = .057$). Partisan identity did not significantly interact with age ($b = -0.01$, $SE = 0.03$, 95% CI [-0.08, 0.06], $p = .788$).

Considering the c' score, there was a main effect of identity, with Republicans having a lower sharing threshold than Democrats ($b = -0.09$, $SE = 0.03$, 95% CI [-0.15, -0.02], $p = .007$). Unlike Study 1, the three-way interaction between age, congruence, and identity was not significant ($b = 0.05$, $SE = 0.04$, 95% CI [-0.03, 0.13], $p = .249$). Thus, the key effect of age increasing partisan bias emerged similarly for Republicans and Democrats.

Strength of Political Identity

We next explored whether partisan strength would moderate the key Age $\times$ Congruence interaction. For discrimination sensitivity (d'), there was a two-way interaction between type of outcome and partisan strength ($b = -0.19$, $SE = 0.07$, 95% CI [-0.34, -0.05], $p = .007$). Partisan strength did not significantly predict sharing discernment ($b = 0.07$, $SE = 0.04$, 95% CI [-0.01, 0.16], $p = .099$). It did, however, predict lower accuracy discernment ($b = -0.12$, $SE = 0.06$, 95% CI [-0.24, -0.01], $p = .036$). Partisan strength did not interact with any of the effects of age.

Considering response thresholds (c'), we examined whether the larger partisan bias for older adults is moderated by partisan strength. An interaction between congruence and partisan strength suggests strong partisans display stronger partisan bias than leaning partisans ($b = -0.10$, $SE = 0.04$, 95% CI [-0.18, -0.02], $p = .020$). Unlike Study 1, however, partisan strength did not moderate the two-way interaction between age and congruence ($b = -0.07$, $SE = 0.04$, 95% CI [-0.15, 0.02], $p = .125$). Study 2 did not provide evidence of greater partisan bias for sharing and assessing accuracy among older adults with strong versus weak party identification.

Discussion

Older adults displayed greater partisan bias than their younger counterparts when evaluating news headlines and reporting whether they would share them. These crucial age-related differences in partisan bias were consistent across both accuracy judgments and sharing intentions and illuminate the nature of age-related differences in information sharing on social media. Study 2 did not find evidence that older adults exhibit reduced discernment between true and false information or that they exhibit lower thresholds for judging accuracy or deciding to share content.

Study 2 tested the effects of intuitive versus analytical thinking by experimentally manipulating thinking style. Compared to intuitive thinking, the analytical thinking condition somewhat increased discrimination for accuracy judgments but not sharing intentions. The analytical versus intuitive thinking manipulation also decreased partisan bias, especially for accuracy assessments. Thus, temporarily manipulated thinking styles increased accuracy discernment and reduced partisan bias. They did not, however, reduce the age-related increase in partisan bias. These results suggest that while thinking

style influences accuracy assessments, it does not explain the greater partisan bias observed among older adults.

General Discussion

The present studies provide clarifying information about the nature of partisan misinformation sharing among older adults. Previous work showed that age predicts misinformation sharing, but the nature of this trend remained unclear (Brashier & Schacter, 2020; Grinberg et al., 2019; Guess et al., 2019). Through an SDT approach, we examined the nature of age-related information evaluation and sharing. Our results suggest that, at least in the contexts of U.S. and Brazilian elections, older (vs. younger) adults exhibit stronger tendencies for believing in and sharing information that aligns with their political identities.

Implications

One possible explanation for why older adults share misinformation is that they are less discriminating between true from false information due to a presumably decreased analytical thinking (Brashier & Schacter, 2020; Grinberg et al., 2019). Our study found no evidence for a link between age and differentiation in sharing true from false news. Nor were older people generally more likely to report greater intentions to share information. These results for sharing intentions extend previous findings that age is either positively associated with news' accuracy judgments or not associated with it (Allcott & Gentzkow, 2017; Pehlivanoglu et al., 2022; Sultan et al., 2024). These findings fail to provide evidence for an explanation based on age-related declines in analytical thinking. Instead, older (vs. younger) adults reported a higher intention to share identity-congruent versus incongruent headlines, regardless of veracity.

These findings may have implications for reducing the spread of political misinformation. In particular, our results question the utility of interventions that seek to reduce misinformation sharing among older adults by increasing their discrimination of true from false information. Increasing discrimination of true from false information is inherently valuable, but it may not reduce age-related tendencies to share misinformation if older adults' sharing of misinformation is due to partisanship, not truth discernment. Instead, interventions might encourage people to behave in a less partisan manner, perhaps by highlighting the value of avoiding information "echo chambers" (Giotakos, 2022; Grant et al., 2024; Romer & Jamieson, 2020, 2021). Interventions like intergroup contact, common national identity, and corrections of group misperceptions can also be promising by reducing partisan animosity and resulting partisan information sharing (Voelkel et al., 2024). Importantly, our findings emerged in both the United States and Brazil. By documenting similar age-related patterns of partisan sharing in both countries, our results suggest that partisan dynamics in age-related information sharing is consistent across different institutional designs (Flores et al., 2022).

Although the focus of this research is on explaining the nature of age-related information evaluation and sharing, our results also inform current debates in the literature regarding misinformation detection and sharing. For example, while some scholars argue that analytical thinking improves discernment (Batailler et al., 2022; Pennycook & Rand, 2019), other scholars suggest that superior analytical tendencies increase partisan bias (Kahan, 2013; Kahan et al., 2017; Sultan et al., 2022). These are not necessarily competing

theoretical accounts (see Gawronski et al., 2023). Our results provide mixed evidence for the role of analytical thinking, depending on whether it is measured as an individual difference or experimentally manipulated. In Study 1, NFC was associated with larger partisan bias, whereas, in Study 2, experimentally manipulated analytical (vs. intuitive) thinking reduced partisan bias. These results suggest that individual differences in analytical thinking might exacerbate partisan bias even as transient nudges to think more analytically reduce partisan bias. This possibility is an important question for future research.

Future Research

Future research might also seek to better understand the explanations for increased partisan bias in older adults' misinformation sharing. Our studies included measures of analytical thinking but did not find evidence that these explained the key effect of age. Nor was education, which is higher for older adults and might be an (imperfect) indicator of political knowledge, a plausible explanation for older adults' tendency to share partisan information. Although mediation analyses inherently carry important limitations, particularly in correlational designs such as ours, our results examined but did not provide evidence for analytical tendencies explaining age-related information sharing.

In addition to analytical tendencies, one possibility is that older adults are more connected to networks of strong partisans (Ghosh & Iyengar, 2019), leading to social reinforcement that encourages partisan sharing. Additionally, older adults' relatively high political activity and civic engagement (Hartig et al., 2023; Quintelier, 2007) may amplify partisan tendencies. Older adults might also have stronger emotional responses to partisan information, experiencing increased affective polarization (Phillips, 2022), which could drive partisan sharing. Testing these social, behavioral, and emotional factors remains a task for future work.

Future research could also explore further the relationships between age, partisanship, and sharing information versus assessing information accuracy. Sharing decisions do not always align with accuracy assessments (Pennycook et al., 2021; Pennycook & Rand, 2021). In Study 2, where participants indicated whether headlines were true and whether they would share them corroborates this assertion. People displayed a greater discrimination sensitivity for accuracy judgments than sharing intentions, suggesting they might share news they consider to be false or not share news they think is true. Further, we also provided evidence that the analytical (vs. intuitive) manipulation amplified this effect. Although research has started to identify who are the people who knowingly share misinformation online (e.g., Littrell et al., 2023), future work could extend these investigations to examine why analytical thinking exacerbates sharing versus accuracy differences. Despite this discrepancy, Study 2 replicated the key interaction between age and congruence in predicting response thresholds for both accuracy assessments and sharing intentions. This casts doubt on the possibility that age-related partisan sharing is due entirely to heightened pressures for partisan conformity, which might imply a larger effect of age-related partisan bias on sharing than accuracy.

Relatedly, our analysis relies on SDT, a framework that has been frequently applied to understand accuracy judgments, but less so to

understand sharing decisions (for an exception, see Gawronski et al., 2023). This discrepancy is likely because the SDT framework is based on a person's ability to distinguish signals (truth) from noise (falsehood). In the context of accuracy judgments, it is straightforward that correct categorizations involve saying that true information is true and false information is false, and incorrect categorizations involve indicating false information is true and true information is false. In the context of sharing decisions, however, interpretation is less precise. So-called "hits," when true information is shared, could simply reflect that people share true information, or there could be many additional factors that influence information sharing, such as engagement, emotionality, identity protection. Likewise, "misses," when people do not share true information, could reflect that people mistakenly thought the information was false, or they recognized the information as true but had other reasons for not sharing it, such as the possibility that the information is not interesting or due to a reluctance to share information that reflects poorly on their political group. While these are important considerations when interpreting d' and c' scores for sharing intentions, it should also be noted that we observed the same patterns of age-related partisan bias in Study 2 for both accuracy assessments and information sharing, suggesting that the underlying processes are closely related.

Constraints on Generality and Limitations

Given the exploratory nature of our studies, we encourage future research to replicate our findings using preregistered studies. Further, the study's stimuli were limited to 12 headlines, due partly to time constraints, resulting in just three headlines per subcategory. While based on genuine headlines, the stimuli were not systematically sampled from the real world (Kihlstrom, 2021). Relatedly, we used an equal distribution of true and false news, as in previous research (Gawronski et al., 2023; Pennycook et al., 2021; Rathje et al., 2023), but misinformation is relatively uncommon, yet still problematic, in naturalistic settings. Our surveys measured sharing intentions but not actual sharing behavior. Our samples may not be representative of people who share content on social media—in fact, few people actually share on social media (McClain, 2021). Future studies should include measures of typical social media behaviors to examine this issue or directly track actual sharing behavior.

Additionally, a careful analysis of ratings of individual headlines reveals differences for some headlines in each condition with respect to how they were intended and how they were rated in the pilot study. Our analyses treated the headlines as they were intended, not how they were rated in the pilot study, following intent-to-treat principles. In analyses that account for headline-specific effects, however, the key interaction between age and identity congruence also emerges (see Supplemental Material D, Tables), suggesting that this effect was not due to any headline-specific differences.

Although our results demonstrate similar age-related effects in Brazil and the United States, country differences in response thresholds and partisan bias emerged. U.S. participants displayed a higher threshold for sharing information and a lower partisan bias than Brazilian participants. Future research might investigate whether this is attributable to different sharing habits, the specific

stimuli used (which were different across countries), or other factors such as two-party versus multiparty systems.

More broadly, future investigations might expand to include additional countries to enhance generalizability and examine cultural, political, social, and economic factors influencing misinformation sharing. Additionally, although Study 1 replicated the effects in Brazil, due to operational challenges, we could not conduct a pretest in the country testing perceived political slant of the stimuli. Given that the key age-related results arose in the United States across both studies, we assume Brazilians did perceive the expected political slant in the stimuli. Future studies might also investigate if these results hold during periods when elections and political identities are less prominent, as the present study was conducted shortly after elections, in 2022 and 2023, with heightened salience of political identities (Grant et al., 2021).

Conclusion

Our research provides insights into the relationship between age and misinformation sharing. It extends previous findings that cast doubt on a reduced discernment explanation and, critically, emphasizes the role of partisan bias. These findings deepen our understanding of why older adults are particularly susceptible to sharing misinformation on social networks.

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Received May 15, 2024

Revision received September 1, 2025

Accepted September 12, 2025 ■